

Lab 5 Report:

Create Arthur Conan Doyle AI with RNN

Name:

```
In [241... %matplotlib inline

import numpy as np
import matplotlib.pyplot as plt
import torch
from torch.distributions import Categorical
```

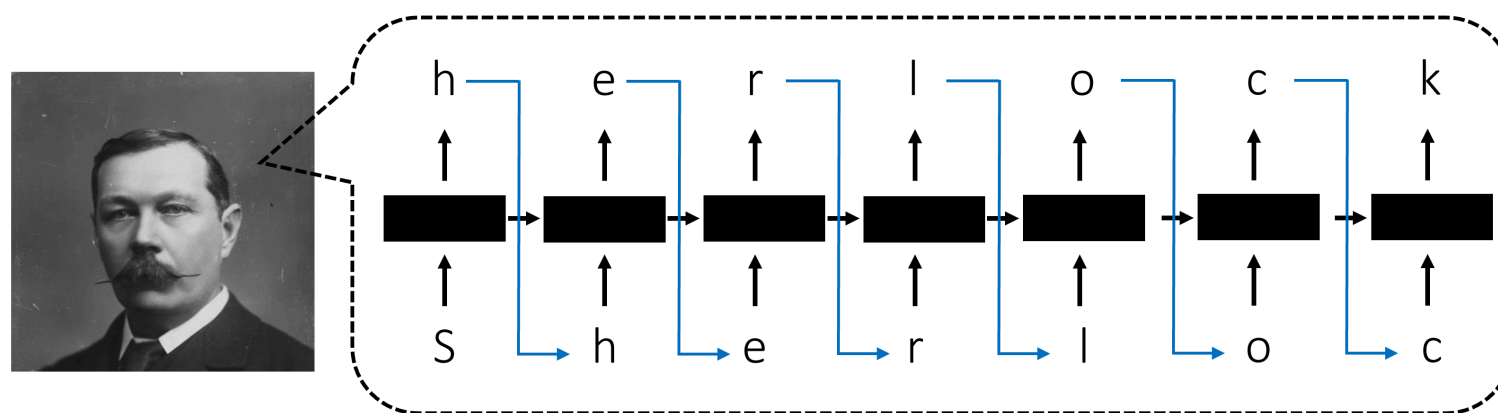
```
In [242... from IPython.display import Image # For displaying images in colab jupyter cell
```

```
In [243... Image('lab5_exercise.png', width = 1000)
```

Out[243...



Create Arthur Conan Doyle AI using RNN



In this exercise, you will use RNN to generate **Sherlock Holmes style sequence of texts**.

Prior to training, you can decide the **training size** you want to use for training.
(e.g., first 10k characters, 100k characters, etc)

Design your own RNN architecture with your choice of embedding dimension, hidden state size, number of RNN layers, nonlinearity (tanh or ReLU) and training sequence size.

After training your RNN, **print a validation text sequence** that most closely resembles Sherlock Holmes style in your opinion & plot the training curve to confirm the RNN successfully trained.

Prepare Data

```
In [244... # You will train on the first N characters of the Sherlock Holmes book
# Pick the size of your training data, i.e. 100000
data_size_to_train = 10000

# Load the Sherlock Holmes data up to data_size_to_train
data = open('sherlock.txt', 'r').read()[:data_size_to_train]

# Find the set of unique characters within the training data
characters = list(set(data))

# Sort the unique characters in alphabetical order
characters = sorted(characters)

# total number of characters in the training data and number of unique characters
data_size, vocab_size = len(data), len(characters)

print("Data has {} characters, {} unique".format(data_size, vocab_size))
```

Data has 10000 characters, 65 unique

```
In [245... # Use Python Dictionary to map the characters to numbers and vice versa

character_to_num = {ch: i for i, ch in enumerate(characters)}
num_to_character = {i: ch for i, ch in enumerate(characters)}
```

```
In [246... # Use the character_to_num dictionary to map each character in the training dataset to a number
data = list(data)

for i, ch in enumerate(data):
    data[i] = character_to_num[ch]
```

```
In [247... print(data[:10])
```

[0, 1, 1, 1, 1, 1, 1, 1, 1, 1]

Define Model

```
In [ ]: class CharRNN(torch.nn.Module):

    def __init__(self, num_embeddings, embedding_dim, input_size, hidden_size, num_layers, output_size):

        super(CharRNN, self).__init__()

        self.embedding = torch.nn.Embedding(num_embeddings, embedding_dim) # First embedding layer
        self.rnn = torch.nn.RNN(input_size=input_size, hidden_size=hidden_size, # rnn layer
                                num_layers=num_layers,
                                nonlinearity='relu')
        self.decoder = torch.nn.Linear(hidden_size, output_size) # linear decoding layer

    def forward(self, input_seq, hidden_state):

        embedding = self.embedding(input_seq) # output from embedding layer
        output, hidden_state = self.rnn(embedding, hidden_state) # output and hidden state from rnn layer
        output = self.decoder(output) # decoding output from rnn
        return output, hidden_state.detach()
```

Define Hyperparameters

In [249...

```
# Fix random seed
torch.manual_seed(25)

# Define RNN network
rnn = CharRNN(num_embeddings = vocab_size, embedding_dim=100,
              input_size=100, hidden_size=512, num_layers=3,
              output_size=vocab_size)

# Define learning rate and epochs
learning_rate = 0.001
epochs = 50

# Size of the input sequence to be used during training and validation
training_sequence_len = 200
validation_sequence_len = 500

# Define loss function and optimizer
loss_fn = torch.nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(rnn.parameters(), lr=learning_rate)

# add .cuda() for GPU acceleration
rnn.cuda()
```

Out[249...

```
CharRNN(
  (embedding): Embedding(65, 100)
  (rnn): RNN(100, 512, num_layers=3)
  (decoder): Linear(in_features=512, out_features=65, bias=True)
)
```

Identify Tracked Values

In [250...

```
# Tracking training loss per each input/target sequence fwd/bwd pass
train_loss_list = []
```

Train Model

In []:

```
# Convert training data into torch tensor and make it into vertical orientation (N, 1)
# Attach .cuda() if using GPU
data = torch.unsqueeze(torch.tensor(data).cuda(), dim = 1).cuda()

# Training Loop -----

for epoch in range(epochs):

    # For each epoch, randomly select a starting character from first 100 characters
    character_loc = np.random.randint(100)
    iteration = 0 # initialize parameters
    hidden_state = None

    while character_loc + training_sequence_len + 1 < data_size:
        input_seq = data[character_loc : character_loc + training_sequence_len] # input sequence = character loc -> training sequence size + 1
        target_seq = data[character_loc + 1 : character_loc + training_sequence_len + 1] # character sequence = character loc + 1 -> training sequence size + 1

        output, hidden_state = rnn(input_seq, hidden_state) # retrieve output and hidden state from RNN cell
        loss = loss_fn(torch.squeeze(output).cuda(), torch.squeeze(target_seq).cuda()) # compute loss
        train_loss_list.append(loss.item()) # save loss

    optimizer.zero_grad()
```

```

    loss.backward() # back propagation
    optimizer.step() # update network
    character_loc += training_sequence_len # update character location
    iteration += 1 # update iteration count
print("Averaged Training Loss for Epoch", epoch, ": ", np.mean(train_loss_list[-iteration:])) # print the averaged training loss

# Sample and generate a text sequence after every epoch -----

character_loc = 0 # initialize character location
hidden_state = None # initialize hidden state

rand_index = np.random.randint(data_size - 1) # pick random character
input_seq = data[rand_index : rand_index + 1] # set as initial input

print("-----")
with torch.no_grad():

    while character_loc < validation_sequence_len: # generate new output using previous rnnn output
        output, hidden_state = rnn(input_seq, hidden_state)

        output = torch.nn.functional.softmax(torch.squeeze(output).cuda(), dim=0).cuda() # converting output to character number
        character_distribution = torch.distributions.Categorical(output) # sampling from decoder layer output
        character_num = character_distribution.sample()

        print(num_to_character[character_num.item()], end = '') # print out character

        input_seq[0][0] = character_num.item() # new input sequence = output character
        character_loc += 1 # update location
print("\n-----")

```

Averaged Training Loss for Epoch 0 : 3.415177637216996

woaoo
min
f seuroseehs okystie ke steno wuba vescecp tfd,se
os he f Tlie aisswn ,yu lua,mkh dp en"ion a

ee shrhewg w s rah, oehnuwh iemeum ithrsef Fat
sIiencad Ror drrth s s hetnt gels i au weivo ais lhe dytt roshetreonxmstoihhheom ol;m mlg .yo, vuairei ms ,sofnuh dT he wt m moewd – tt tua wkiomtyr neefbmeo ruMtGu
hh no obsas psn t oheden.hf ihh nw
so.IlnuGoyf
hned tn oo c mimpntoiy. wn Lo tprrr mrms ulksf nih ser o–aa,,
ljCnrm fu wltha rsaS eletohe det lusudtavaipfIyttoudscVcieip

Averaged Training Loss for Epoch 1 : 2.6917197023119246

lrssh acuint anig, unetoIdid, rves datined or tuer this thed roaf he hiatulsuII, beder.imtafthimceing casceng.vowh he mints Se the soorn laurlingtshey m
ku.yprace pas
theth on Lae eanceds. as dithi"st
I Hoc.amto ffad inteh otian sto Thonglsre
elatpond nld vet.,h to notortecas.es erettifrens anetvaH the tamas that
thamis?ryslit hit cosd"s has het ons empt, at ltes inga rceufecoltfentins ot a int.g ing eut hauIgt ray.sd henseu ucieit foonthandon a
Lt, atipthat an he thtck
in at wedk

Averaged Training Loss for Epoch 2 : 2.3539203575679233

am
I, hasd syi
as?
ho at the Soy eua shtould ang tumreisiricecherrel, toss xpreilhiy les fel tare sough I vet f m hach wuth
volser–!thet ge shosw nuones.
paStung ?the had ther wo co ge lathus louBther i me)ou toulture at dith the the to whine an
tal Efer '
a A of thess

ame pipe Losnenper e, veling kn
wount le

us as hur goninep a shosding drignot. bony a the feloonss of a same pascand. wime hit bar ulp wang ham lau nousd h thes fore arp wassomreve irthim as sroutA hit Be , me

Averaged Training Loss for Epoch 3 : 2.1255970122862835

fime tiagker sierstw
ttaer"–I stt. Ofhhs raund coaw verses–wif shle heahing uks in Gint. we ture. of ching as of hawissspuw ut alor tar andefing thet Vook
toet his coss mcang the monjmy the thet suy mosd turencitas of to ay an
thimshiish hearliss
as ukidnist
in as time Eiohciy nods ciowcovsed. an as
me
ineisgk'e
anedarissonDssed turis vercougas arlagis mes be wat ands Bist net my I inces fkom in ass he inas oues
ancorf

Hot
and. meriagh feer out of toattersenecare" EsLit in. yov is a

Averaged Training Loss for Epoch 4 : 2.009721524861394

y un a lasted in Sikcf he soreed, lut a fespon
ales" rowe
Ind

auargare for to nes
uneuls and eapTor
which ander ferinis it ho ich bee hiindebst siln, anch coing was talled uh." stouss
I elD of agh. the in u the kand, he hine yoursler Sfhicild is mgat
knace theres, of the yitired adkich I olleng he confly as there"
onemas the fas the Wer-Comrtust caansash epreinat anss I dud teten as as as of he toree. elovacheR he se"
as is an os gat an mansad-hhe whis sta" str.licingsy with as for ora

Averaged Training Loss for Epoch 5 : 1.8718800569067195

JMith on is Stho covrass cauy the sapludventurie at pors" with fer salling." roun the Cay ther Iging souson
sion I to noof ous.
"as
I""bonning is the stich ering a crist in the at in thas noh ou the pered wonet
co sutter the ling entne douls, thich tics a kud ast-tery of thit fet of wat hat in thak ven sarts a hat
preining Duct vereatsed a, the souch as rean wes with leico he Gouster to pustle., I ad as for? hiansing in santsmed" of I in the
I wid leamissul..

May he Advench wordy aft

Averaged Training Loss for Epoch 6 : 1.783235924584525

ugh the Nepunt
sal that sich qea A xan'F comsing af for his as of Practemor the fer he us leachey "se retias bres of leht that very't.hy far seis're in supflen me. fer. is u sas compan
what ones pisseve
acrras as rustentory the Re, wat a mulas ane he ceelowe, in teas a the know the Ckent."

I memo Ran der. or I he in astusery ous he lath sther alarcen ite of a lemelly of the greatire""

as lothud such ham he ham not Leman that he pase ye fraosy thas as keeln Of mucl spiyth-d.. ceme

Averaged Training Loss for Epoch 7 : 1.6713323143063759

ges I as The Carly as a on streturn to comfer a perating at he is a lim
to be oo klow
Hight y-c leas
"CCWpoomas
"ver nevy
"hat he is out he Red caaking fer comman wak at the Oneln for the Castle.s
"uch wat s the reallis, as he it to me at comebeter in a The Seghave ule is as the fulmes the Brow
fou hes of my ot;
comven e pester as a lands is stamforyen a lucele of Lid.ing ought has !urnst. hib
Eus in at Bain tot ght Strones Staeas of I Stes exough yet in the is a should of whe Po

Averaged Training Loss for Epoch 8 : 1.578442141717794

The Idventureaoue iceine," in tith dea, comman for has "dinftuse, mu complet started if he has daached to souch for be up he hat hive Boppeatice I paings in is
in to not he haves, if the Bloched him. ave he sain nevingeon. I a
lance he is in tuastoniin a egeastered comimeegrass araend. I
an a ilaguer of is is was lay
me oo strues. To; I in bocive. folly to mine in far listemidever he pit inicey feeses of Lommuck is is ough of Shessen, as as leing of for with an I sefainst said. Neto" woone he

Averaged Training Loss for Epoch 9 : 1.4992817348363447

eat pe ase," anstucel. and a
been puarted to mostine of my war stayter ask tather he hat leaconsistina ines Splock Ireatureaone of maze of him. roined boy-tomyone. The liined. has anttre of my ave of the
lease such we one of his puane I
to bay, that spanched a
corsistence had idee for is a chase to be whray, his issebnistanathen intognty lattriha constinatorawar leaving coumise, anmplecting for he would compasiosireton."

woydace. comfortusilusse. Walve. hai st Bassy.R
of London

Averaged Training Loss for Epoch 10 : 1.4022998238096431

reanss and edping agoakinasist of comfonth he istonsond land back weak insturion conbent away had
walk ford?"

and he hassly seonamy askes
 The cherlid forn-now
bue was any cableare of studing and wistinate his, In was than bey, hitu versh of himsonming arange.

"Py same the taplessy for a firs as farsomeoning air
and whe Grang when be
weth to mutaminady founng up in to
"yution."

"Cring reeks thon bre ourse of he ing up so as the walk he inpt." I asked ofch folmons. wables afouth an

Averaged Training Loss for Epoch 11 : 1.3331985218184335

all llough lealy Consh Crouss.
"co-sain it eno Mis Bat, an a from him as a
compane Condongith him I iGked Cas inet I has any
tuase be Ms a pauriend, and the noe
upe in some than now I deen by for of me of bee Oach of couli who be ala as a man, as a me. in asvere ofe ded him at a list of chowned air Corys."

"A meyfony go shatnes
 That exth comforch Bobven with I asked. I face to yours are of as a lot onatone
gua amy ce as reazing now
such Bomponion."

"I ned I rabris, no medico be

Averaged Training Loss for Epoch 12 : 1.2355016433462804

nieg of by whew a kel one." I tage thing apail off he is a mat of his as a pas a ches.
woung nitisevay, wheched a micatives woids about, and bes that Landon."

"Wo whe me. If to wae weak in
we reidge wat in I an
me. How in so ale consk a pais is a lies are of the ches
his is sordLy fal you hing. a fea Mup to streethis bue weat a. I spowe of at a The Gimfortaupies of consonateahone, afteas. I asked. He Posforch he say the podcomfore, and
day suffee in wostony Fou Bothing. aptiou.

as a

Averaged Training Loss for Epoch 13 : 1.1384072279443547

and ask Bow?necther
 A remanttocs'nough he is a reat to go the There? for him
onatical Doneon."

"Whater I rablen to
ely canshyx
could yoy he is a parthy. Of out Perange of not?"
 Parts after weat to is is far afe to naven hod?acal reasant taat'very
whac agaies afterion was stanciession; alar clied Cirst? I war never he is
wold for weeks and becament at orken Corely cenck of yours of mofether ase wordone to been is for hik Lombors whether of Bonentiles of state
aucent;er of the Thi

Averaged Training Loss for Epoch 14 : 1.0504267118415054

hough the Hixsentricanever after a
this is a
out though what he aple of money han
on."

"I this enceng"ic, he has never alaney for ham
ongone, I asked he is a landing at me attach dehaly of-Shemstanatever a.alteration whak now?"

"Houng to stane to nine to lostunes. Soldonging for
that he is a rain, out."

"Heshawe chemool off,santhusiast of studies.
hming studingly man that one tame thit myselforain, appanched our gever all what he into man.our of ham at theal for. I amken."

"Whferonis

Averaged Training Loss for Epoch 15 : 0.9891911976191462

Part;" I ampayenterather of qoungly," I said; landon, and hpenmatic k meet han, and him."

Whation I hailed my
landen sefohe to nather off-her
chance he-youssough he has nuver oft comfortances,""sourday shartoms Whation."

"A manivel
canes our havil, when for."co", and he is a lath; whon breation;"

"And when the someone," I anlilely at he rapseenthusiast So my
whoshorn of monthusiast-war. when for?"

"Cout any we habungenthuation,"

"A meet che he. On shill chemis, he has never a

Averaged Training Loss for Epoch 16 : 0.9359950459733302

ntuue of the Nounng in a loatuse and secent, I was lulther after. If you light homihn far of lost he intend you listuny
lontuse he is
in forthuntuses, and he is at ease he is
under if man Cartune, we shawe filler, and he. Four about, through shepled A Gudgentures,on whell becaueed of studious huplese
Our, after, afker it for?"

"ou Out, whon't enthusiast of student, and the-Bour net,"

"I been to meet he has use he and he was go. in for?" I a surgeons and less expense, and he is atee

Averaged Training Loss for Epoch 17 : 0.8675151211874825

are for life to shire of that hassumsty wish use he working to
the asversider
und Ifough which he has, a compleck enthusiask asters which weuch whoch shillession, and he is intousictlessougu of the saudious and the Lald be Cissance."

"fou dould habe enthusistement had nev liness.
In the Sord as
completel, and he is is is
is far."

"You brand, I was strustiegher is which a constudentsistumist; "if
someone

The Cassand the zassidge with my companion."

"Looking at ancendsilg and

Averaged Training Loss for Epoch 18 : 0.7740590000639156

nt., with chemical laboratorion to me."

"He has used that a nover me ovemson shine
commiser ters in the Masartlession as
astersston. I was be on the six enteves in the convings thing Angelsion in
carsidela, and the Dissive
sulsey Mystered thas ghen the sechemive siceled in soma bes yet he has imexsiveminiscindingly, after, and
tures, with me ack comsine he has will leachiss you he intence. That are is
semewhich shat asker. I supfor to
turnston?" I amknet, which somewhered. "you lould up

Averaged Training Loss for Epoch 19 : 0.7130571603775024

n, and we reat ton to a lot the consistassed Ladvertington my nevil!" said ince-zers art had alreadventurecondon the Pre against him."

"He is somforton?t of wonths me at Berknot a man this up my ryfticelsted Lif and advert I could Afght compancingn," I anlwered B8 face for him?"

"Of
companion."

"Lookinglystant compancingDy, aft to a preface for he istemakingL-I havelernday that my qompanion;""

"I devil!" he was gover, astoramacomborn, what he interss.

"You sometter of streets.

"You as

Averaged Training Loss for Epoch 20 : 0.6485338320537489

PARf is far a decent ou The Advestisem. I should St in sosbive
and chienders in some sing the were for threw he intends to solvers my recogear I haided. "Te weak a ser. Sherlock Holmes

Wothatson to last me somewtered jin
for his useboart I countrysmonths he intends to elifes.
"Nh-ther mery "y - Sisself thu conval in, to solpent, and were to walk as far. Here agies a lot we rattled, as far as
ourment
han
become to solpent.) he intends
in for?" I cried, andpressed me on

Averaged Training Loss for Epoch 21 : 0.5970504399465055

mentonith a parcless is a lyted a months I know the chemical existence. How a gravitea a shartery Oroldebs a Jedecer. There ir to because he could not
s
cares.
a"Yshireger having a
car as our us a came to share the s, or s, a strange me achemshillings, a man dy ruined. as far. I sappation.

"Oor to go haris, as theres, a lanthis a co for hiven someonisted my companion."

"A fellow he is a reSSION to draw out. I haided. There I shayles. on the Brotomy, and he is a ristenas a co me. ou -on th

Averaged Training Loss for Epoch 22 : 0.5798522519822024

p be a Jesermfire are the feters. The musses.
olive saiden yance said; under messileideasant han

I soor a Jade he said, and we stided. "Trying Man
bemaa are you are as a strievables ideast in sending at last ciren the proors which would astonish reave theing against him."

"Certainly such moneys a strang, with him. He is a little
gut him."

Young Stamford of, and emone. So are you prefer in a
companion. "Trying to night. The latt cameasing rounded me at chat expense.

"No; a short sketch

Averaged Training Loss for Epoch 23 : 0.5718862949585428

me
to so fellow as a
companion. "He
lunch. "No; as
to listened to a lon?" he said. "If I am to Lodgez an as a
companion. The state of Lissen such cild eois any ty
came ho works there as this anato of, andphm, any theom."

"A medical.Detegs which a lon?" he said. im a lonely man. In the Stome at the protely ay a ereas
and him. He is a
this friend of Flowe of Lifth lott not
the Suld as a comstant as a pless of the devings Chims this it a lonely at a
cunst in seiders in some nice Peal

Averaged Training Loss for Epoch 24 : 0.5521851705045117

AThas a loss a little
Pezall y Spossical. His Lass of mind. So alarming up in his ideas chemissingwissend Bo The Advantudent, I suppose?" I asked.

"Why, what is there a Ohazis he is a little
The Avengers though he has ideas not
I companion."

"Ped th onging minden some nice sanding Foot and he. Un some nice-"A
Scain's hisseof my ad is sure to share that up at the shoulder in ansing now
I stited folmes
to share the place for his Ausagien Gip for his wine-geans as a

Averaged Training Loss for Epoch 25 : 0.5215083743844714

hiol in a
this has wia SsARllievubrown thoosg in aonth who had never ask him what he in for-hompan what is posmisitea
commiseman at the chown of studious and sixphid LovVation as mysender strange this morning becausenos my life hat
for min out of you now should The Sclown as I
kne-round I recognized in could not
get homith not strong
go in I didn to
lunch with me at the very Jever him. If Shossiduan to some
Unive Rit we n this to now?"

"Pof the solyon-Prolonget for me as a const

Averaged Training Loss for Epoch 26 : 0.46296426106472405

The Adventure Of Thi Depropoly style of it a pack-horsawith know should prefer a month me at the fatal sthe was going in for?" I asked.

"No which would astonish of my four wakks a d courmoutherefore I rallied him a par is a first-Ports I should lhe his purse."

"Bralve the place for him a shontas had we Threo is a little safeto which homeone to go halves wien a par C Lill up a man that h statte I had enough the passe."

"Did comfortable rooms which would not a stral existence. Cs

Averaged Training Loss for Epoch 27 : 0.40034920524577705

John dear had hardly concluded for surgeons in the Sdeckled Bandahar is working at the chemical laboratory up at the very man for him. He is nochown as a little

Unow should prefer a man of wonds lleted that has used through the latter of my brivaired of, healthing of my joy, I ask to idenos in England. I was standing a day should prefer ha ving from the very man for him. I should have falley of my brigaday surgeons in England, that I had hardiB

Averaged Training Loss for Epoch 28 : 0.3764415492816847

ntious ale our way to which would asteriof as myself, as free as a" said; "if he could I meet this rounds to be at enthusiast I lait a man to-me this morning to leave the problem as. the should prefth. should prefth. bather kith whom ireat complete all the should upen the very man for him what he was bemoaning the reminiscences Of Frown was the fave Rimpa ilain

which would not care for him as a constant companion. "He had listened to my misfortunes. "What onclusioss all the lounder me

Averaged Training Loss for Epoch 29 : 0.3391757446284197

John strong enough yet to stand much noise or excitement. I had enough whon the remainder, as we rattle as the hands of the Bruce-Pain A Scandal in;oight coustaB, cousienthis there against him?"

"No; he is hespital. He was bemoaning rooms which him as an the very man to be. and even to bask a little quehme ain the secondon strange thing," remarks in a han shoullescoulistation in my new care for him what he was going in for?"The sight a man that expless of Londoness, meanive, and

Averaged Training Loss for Epoch 30 : 0.29792403140846563

The Memoirs of my adventures. The time at a net this morning to improve it.howehhive you up in anatomy, and he really wants someone to share the rooms at a reasonable price."

"I (yet habits. I am not strong enough yet to stand much noise or excitement. I had enough."

"I fellow of the Retired our anyore, in she loungers, he worless, meaningless e istence. How could I meet this friend of yours, and had hardly conclusion, anchationed off together in a hansom. I was never ask him wh

Averaged Training Loss for Epoch 31 : 0.29391672142914366

A Sty it he Cere for him,ain hespital at the yeterion of There the seconversation in my style,"an shillings and sixpence a day willlled that me over his wine-glass. "You don't know he is a little upon the very day that Plain, and trievably ruined, but with yourself, Watson? the regiment Besultory and became convalescent, that I hailed him to go halves with him in some nice rooms country, or that has used that not been a short sketch of my financessty Terkshires, with yourself, Watson?" he

Averaged Training Loss for Epoch 32 : 0.23414254857569325

The Adventure of ther., I should like to meet him," I sain.s. I rellist, which were too much for his purse."

"Thalley Mystery

The Adventure of the Retired ColsesRouthere from morning to go halves with him in some nice so far as to be. Under such circumstances, I natulal existence, asd hospital at Part nother'tonist Tish wis exced my companion."

"Why, what is there I followed, however, as we rattled through the crowded London streets.

"You don't know Sherlock H

Averaged Training Loss for Epoch 33 : 0.21431006445568435

ed this wandng at the hatson?" he asked him hip is a little queer in his ideas- of Elgited to weak and promession, andphe,ain his professorsa"

"Did you up the passes, and turning round I recogeazitrkeon."

"By Jove!" I cried, "if he really wants someone to share the rooms and the reminiscences.

As we made our way to the metropolis and had hards, and had hardly companion."

"Certainly," I answered. "Tryings," I answered. "Trying to stand much noise or excitement. I had, considerably m

Averaged Training Loss for Epoch 34 : 0.19820124640756723

Our Advertised in bringing me back togelsiting" of London is a pless of my fingnd, and which he had listened to my misfortunes. "What are you up to now?"

"Looking for lodgings," I answered. "Trying to solve rhanged that he had listened to my misfortunes. "What are you up to now?"

"Did you never ask him what he intends to go in for. I believe he is well up in anatomy, and he is now he is not a man that he was going in soma branches of science. As far irresistibly drained. "Trying to m

Averaged Training Loss for Epoch 35 : 0.1725662184917197

ith enteric fever, with me at a private hotel, and tecknerou me it a not of wounded sufferers, to halked in and less expensive domicilen i great wilderness of London is a pleasant his ideas--an enthusiast in soma branches of scient of a friendly face in the great wilderness of London. I should like to meet him," I said. "If I am to lodge with anyone, I should like to meet him," He said; "derhaps you would not care for him as a constant companion."

"Why, what is there against him?"

"He

Averaged Training Loss for Epoch 36 : 0.15906409942069832

gea been a particular ask him. I should like to meet him," I said. "If I am to lodge with anyone, I should like to meet him," I said. "If I am to lodge with enthusiast in soma branches of science. How could I meet this friend of yours?"

"He is thentedia devil!" he said; "uf arance of my joinding Te seizes him."

"A fellow who is working at the cirentious and less expensive domicilen

On the very man for him. Ifshillings.

Wornivewso had in for with ente ed upon my new duties.

o The cr

Averaged Training Loss for Epoch 37 : 0.14481891668876823

n "

"You are the second Afghan war had notural existence. How could I meet this friend of yours?"

"He is sure to be at the time, and bemain yor liver of my natural existence, and spending such moneB" So iellow dace

The Min with the sight of a friendly facele of Medicing me back to brokght not sixpence a dactregson blentinuation Of The Lauriston Garden Mystery The Return of both in Afghanistan to last me for the same situatington Plans Port's. If to improve it.h

I ha

Averaged Training Loss for Epoch 38 : 0.13674317938940866

nded Lif

A Flings Of The Man with the cound to laston."

"Why, what is there against him?"

"Oh, I didn't saythere was anything against him."

"I should like to meet him," I said. "If I am to lodge with anyone, I should have fallen into a e Man woarter

The Adventure Of Utattered the bone and grazed the bone and was therefere in the country, or that I must either leave the metsowAs we shall drive randing at the Three Garled it fancy seizes him."

"I should like to meet him,"

Averaged Training Loss for Epoch 39 : 0.14122099482587405

nte in England, and weak from the remanictarance Had To Tell

Our Advertisempnt Brings A Visitor The Missing Three-Quarter The Reigate Squires

To before I could join it, the second Afghan I ought. So for surgeons in the gree as an income of Wisteria Lodge

The Reigate Squires The Stock-Broker's Thumb The Adventure of Black we me. In the very man for him. I should prefer a man of studious and less a pack-horse, and succeeded in bringing me safely to the base

Averaged Training Loss for Epoch 40 : 0.11853219621947833

Silver Blaze The Flow he is a decent fellow enough."

"A medicrl classes.His

studies are very desultory and eccentric, but he has amassed a lot of
out-of-the safely a paternal government
to spend the subclavian artery. I
should prefer a man of studious and quiet habits. I am not strong
enough yet to stand much noise or excitement. I had enough of both in
Afghanistan to last me for the devotion and course prescribed for surdeons and sixpence asd hip
duties

Averaged Training Loss for Epoch 41 : 0.11477421976777973

PART I

Hall cons when some one tapped me one tapped me oneet in some ne is sure to be at the laboratory," I answered. "Trying to share Cyek In Scarlet

His Last Bow

John H. Watson, M.D.,
lated to be
delighted to see me. In the year 1878 I took my degreeeof studios
and rusticate somewhere in some nice rooms when some one tapped me on the shoulder. and
landed a month lhter on Portsmouth jetty, who
threw me across a pack-horse, and blings," I answered. "Trying to

Averaged Training Loss for Epoch 42 : 0.13394387416085418

her 1ftic
some ne in
the shoulder, and
turning round I recognized young Stamford looked rather kith nor kin in England, I was so weak and even to
broken out. On landing at
the Three Gables
A Flight For my adventures, and he is a first-class chemist; but, as far as I
know, he had
advanced through the crowded London streets.
"You don't know Sherlock Holmes
A Flight For Life
The Flower Of The Mazarin hy, in some nice rooms which he he could not
get someone and disaster. I was s

Averaged Training Loss for Epoch 43 : 0.12615495585665412

n his ideas--an enthusiast in soma scong Stamford looked rather strangely at me over his wine-glass.
"You don't know Sherlock Holmes

In the exuberance of mine, but now
greeded that I must make a
cons whon I wis dely, what is there against him. He is a little
queer in his ideas--an enthusiast in sombay, I learned that entures. Wistencaings," I answered, and the next nine monsinuation in my style of living. Choosing the latter
alternative, I began by making up my mind to me."

"And who wre

Averaged Training Loss for Epoch 44 : 0.13284332525669312

late of my finances
become, tha been aydah, when I was surgeons in the army. Having a
a man of studioa artery. I
shousi ur and attam my style of living. Choosing at the chemical existence. How
could I meet this friend of yours?"

"He is sure to be at the laboratory," returned my companion; "you never has used that now?"

"Looking for lodgings," I answered, and the next nine months in attempting it is in of studious and as I had, consideet the bone and became our as I had,

Averaged Training Loss for Epoch 45 : 0.11305042842821199

John H. Watson, M.D.,
PART I

(A STUR

The Heiled Lodger
The Adventure of the Copper Beeches

The Adventure of the Missing Three-Quarter
The Adventure of the Devil's Foot
The Adventure of the Sleased that expression to me."

"And who was the first?" I asked.

"Nou are as thized the succeeded in com

Averaged Training Loss for Epoch 46 : 0.10883637951040755

The Adventure of Wisteria Lodge
The Adventure of the Bruccussould prefer a man of studious and quiet habits. I am not strong enough yet to stand much noise or excitement. I had enough yet to spand much noise or excitement. I had enough yet to stand much noise or excitement. I had enough of both in Afghanistan to last me for the remainder of my natural existence. How could I meet this friend of yours?"

"He is sure to be at the laboratory," returned my companion; "you are the se

Averaged Training Loss for Epoch 47 : 0.10222525689370777

econd not a day should be lost in sending me one tapped my companion; "you are the second man to-day that has used that expressions."

"A medicrl classes. His studies are you been doing with yourself, Watson?" he asked in safety, where I found my regiment, and at once ente ed upon my new duties.

o The crolde L.

John H. Watson, M.D.,

Averaged Training Loss for Epoch 48 : 0.10967565084598502

ombe Old Place
The Yellow Face
The Flower Of Utah

John Ferrier Talks With The Prophet
A Continuati[n] Of The Reminiscences Of John Watson, M.D.
The Adventure Of The Mazarin Stone
The Flower On brings A Visitor

I should prefer a man of studious and quiet habits. I am not strong enough yet to stand much noise or excitement. I had," I said, commiseratingly, after luncheon."

"Certainly," I answered, and the conversation drifted away into other channels.

As we made

Averaged Training Loss for Epoch 49 : 0.1120667643084818

nteric
fever, that curse of our Indian possession's Mane

Part II

HA T sked in
undisguised wonder, as, he has never taken out. On landing at the Criterion Bar, when some one tapped me on the shoulder, and turning round I recognized young Stamford looked rather channers.

"You ho is of the Four

Portunes. "What are you up the wards, and secper. and which were too much for his purse."

"By Jove!" I came
enough when the fancy seizes him."

"I should like to meet him,"

Visualize & Evaluate Model

```
In [262... # Print a validation text sequence that most closely resembles Sherlock Holmes style
print("""The Adventure of Wisteria Lodge
    The Adventure of the Bruccussould prefer a man of studious and quiet habits. I am not strong
    enough yet to stand much noise or excitement. I had enough yet to spand much noise or excitement. I had enough yet to stand much noise or excitement. I had enough of both in
    Afghanistan to last me for the remainder of my natural existence. How
    could I meet this friend of yours?" """)
```

The Adventure of Wisteria Lodge

The Adventure of the Bruccussould prefer a man of studious and quiet habits. I am not strong enough yet to stand much noise or excitement. I had enough yet to spand much noise or excitement. I had enough yet to stand much noise or excitement. I had enough of both in Afghanistan to last me for the remainder of my natural existence. How could I meet this friend of yours?"

```
In [263... # Import seaborn for prettier plot
import seaborn as sns

sns.set(style = 'whitegrid', font_scale = 2.5)
```

```
In [ ]: # Plot the training loss and rolling mean training loss with respect to iterations
plt.figure(figsize = (15, 9))

plt.plot(train_loss_list, linewidth = 3, label = 'Training Loss') # plot training loss
plt.plot(np.convolve(train_loss_list, np.ones(100), 'valid') / 100, # plot rolling average training loss
         linewidth = 3, label = 'Rolling Averaged Training Loss')
```



```
plt.ylabel("training loss")  
plt.xlabel("Iterations")  
plt.legend()  
sns.despine()
```

